Network Troubleshooting

1. Ping www.youtube.com

Command: Open Command Prompt (Win + R → type cmd) and enter:

ping www.youtube.com

Windows sends four echo requests by default and displays the response time for each successful reply.

Example output format:

Reply from <IP address>: bytes=32 time=25ms TTL=115

Reply from <IP address>: bytes=32 time=28ms TTL=115

Reply from <IP address>: bytes=32 time=24ms TTL=115

Reply from <IP address>: bytes=32 time=27ms TTL=115

These are illustrative values, not actual test results.

Record your results:

Item	**Your Result**
Website	www.youtube.com
Resolved IP address	Write the IP shown
Packets sent / received	Write the totals
Average response time	Write the final average in milliseconds

If ping requests time out, it does not necessarily mean YouTube is down; some servers or networks block ICMP traffic.

2. Trace the Route to www.flipkart.com

Command:

tracert www.flipkart.com

This command shows the network hops between your laptop and the destination. Each hop is usually a router along the route.

Example output format:

Tracing route to www.flipkart.com

1 2 ms 1 ms 2 ms 192.168.1.1

2 10 ms 12 ms 11 ms <ISP router IP>

3 18 ms 20 ms 19 ms <Intermediate router IP>

4 * * * Request timed out.

This is an illustrative example, not the actual route to Flipkart.

Record the intermediate hops shown by your own command:

Hop Number	IP Address / Hostname	Response Time
1	Your router's address	Actual result
2	ISP router, if shown	Actual result
3 onward	Other routers, if shown	Actual result

A timeout (*) means that hop did not respond to the traceroute probe. It does not necessarily mean traffic cannot pass through that router.

3. Find the IP Address of www.zomato.com

Command:

nslookup www.zomato.com

Example output format:

Server: <DNS server name>

Address: <DNS server IP>

Non-authoritative answer:

Name: www.zomato.com

Addresses: <IP address returned>

Record the results:

Item	Result
Domain name	www.zomato.com
DNS server	Write the server shown
Resolved IP address	Write the returned address

DNS may return multiple IP addresses, and the answers can vary by location, time and DNS provider.

4. Ping Another Device on Your Local Network

First, find your laptop's IP address:

ipconfig

Identify your phone's or friend's laptop's local IP address. For example, assume it is 192.168.1.20.

Run:

ping 192.168.1.20

If the device is online: You may receive replies showing the response time.

If the device is offline: You may see Request timed out or Destination host unreachable. However, these errors can also occur because of firewall settings, incorrect IP addresses or network isolation.

Record your actual target IP, replies received and packet loss.

5. Short Report: Which Tool Is Most Useful?

1. Ping tests basic connectivity and measures response time. For example, pinging www.youtube.com helps me check whether the destination responds and how long replies take.
2. Traceroute (tracert on Windows) shows the intermediate hops toward a destination. Running tracert www.flipkart.com helps me investigate where delays or routing problems may occur.
3. Nslookup checks DNS resolution. Running nslookup www.zomato.com shows which IP addresses the DNS server returns.
4. If a website is not loading, I would use ping and tracert to investigate connectivity, while remembering that a website may block ping.
5. If my connection is slow, I would compare ping response times and inspect traceroute for unusually slow hops.
6. If I suspect a DNS problem, I would use nslookup to see whether the domain resolves correctly.

Conclusion: Ping is useful for connectivity and latency, tracert helps investigate routing paths, and nslookup helps diagnose DNS resolution problems.